

LE NGUYEN GIA HUNG

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EDUCATION

FPT University

Bachelor of Engineering in Artificial Intelligence

Ho Chi Minh City, Vietnam

Expected Dec 2027 GPA: 8.62/10.0

- **Honors:** Certificate of Merit: Honorable Student of Trimester (Semesters 3, 4, 5).
- **Relevant Coursework:** Data Structures and Algorithms, Linear Algebra, Introduction to Artificial Intelligence, Deep Learning, Statistical Analysis.

PUBLICATIONS

Gia-Hung Nguyen Le et al., "Proactive Air Quality Forecasting and Health Alert System for Melbourne." *Australasian Joint Conference on Artificial Intelligence (AJCAI 2025)*. Springer, Lecture Notes in Artificial Intelligence (LNAI). Q2 (SJR 2024).
DOI: 10.1007/978-981-95-4969-6_34

RESEARCH & TECHNICAL EXPERIENCE

Proactive Air Quality Forecasting System (AJCAI 2025)

First Author & Corresponding Author

Academic Research

Jun 2025 – Nov 2025

- Designed and implemented **SmokeNet**, a Transformer-based architecture for time-series forecasting of PM_{2.5}.
- Achieved **MAE of 0.7470** and **R^2 of 0.9545**, reducing error by **35.9%** compared to XGBoost baselines on 2024 test data.
- Demonstrated model robustness by outperforming baselines by **57.7%** during the severe May 2024 smoke event backtest.
- Engineered an end-to-end pipeline processing 4+ years of OpenWeather data with GPU-accelerated chronological cross-validation.
- Integrated Explainable AI (SHAP, Integrated Gradients) to provide interpretable health warnings for public safety dashboards.

MERR-GAT: Explainable Multimodal Emotion Recognition

Second Author & Corresponding Author

Academic Research

Oct 2025 – Present

- Developing **MERR-GAT**, a novel hybrid framework fusing **Wav2Vec 2.0** (audio) and **RoBERTa** (text) via Graph Attention Networks (GATv2).
- Designing a dynamic conversational graph where nodes represent utterances and edges capture temporal and speaker interactions.
- Implementing an intrinsic XAI mechanism where learned attention weights provide granular insights into the model's decision-making.
- Benchmarking on **IEMOCAP**, **RAVDESS**, and **MELD** with a dynamic classification head supporting 4, 5, or 6 emotion classes.

End-to-End Vietnamese Speech Recognition System

Co-Author

Academic Research

Sep 2025 – Present

- Developing a parameter-efficient (5.4M) end-to-end ASR model using a **4-layer CNN frontend** and **3-layer Residual BiLSTM encoder**.

- Curated and trained on a massive **745-hour Vietnamese speech corpus**, optimizing with Connectionist Temporal Classification (CTC) loss.
- Achieved a **Word Error Rate (WER) of 33.09%** and **Character Error Rate (CER) of 15.28%** on diverse test sets.
- Benchmarked performance against foundation models (PhoWhisper) to evaluate trade-offs between computational efficiency and accuracy.

PPE Detection System (YOLOv8)

Developer

Course Project
Sep 2025 – Nov 2025

- Built a real-time object detection system using **YOLOv8** to identify 9 classes of personal protective equipment (helmets, vests, gloves).
- Developed a safety logic engine to automatically classify workers as "Compliance Safe" or "Unsafe" based on detected equipment combinations.
- Deployed the model via a **Streamlit** web interface featuring live webcam inference and automated violation reporting.

LEADERSHIP

SpeedyLabX - Student Research Group

Founder & Student Lead

FPT University
2025 – Present

- Founded a student research group of 10 undergraduates; led weekly technical workshops and reading groups.
- Coordinated submission workflow (milestones, reviews, camera-ready) for the group's first AJCAI-accepted paper.

CERTIFICATIONS

- **Data Science Fundamentals with Python and SQL** – IBM
- **Statistics for Data Science with Python** – IBM
- **Introduction to Containers w/ Docker, Kubernetes & OpenShift** – IBM

SKILLS

- **Languages:** Python, SQL (PostgreSQL).
- **ML/DL:** PyTorch, TensorFlow/Keras, Hugging Face Transformers, Scikit-learn, OpenCV.
- **Data & MLOps/Tools:** Pandas, NumPy, Matplotlib, Git, Docker, Linux, Weights & Biases (W&B), Streamlit.
- **Technical writing:** Blog posts and technical documentation (portfolio: hei-portfolio.vercel.app).